NCI Adult Central Nervous System Tumors Treatment Patient Guide

Understanding Brain Tumor Symptoms

Symptoms depend on tumor size, location, and rate of growth. Common symptoms include:
- Persistent or unusual headaches, often severe upon waking in the morning
- Unexplained seizures (epileptic fits) or muscle spasms
- Cognitive changes, memory loss, confusion, or personality shifts
- Progressive weakness, numbness, or paralysis on one side of the face or body (hemiparesis)
- Visual disturbances, including blurred vision, double vision, or loss of peripheral vision
- Nausea, vomiting, and dizziness
- Difficulty with speech (aphasia), language comprehension, or hearing balance problems.

Diagnostic Procedures

1. Neurological Exam: Evaluates reflexes, vision, hearing, balance, coordination, and mental status.
2. Brain MRI with Contrast: The gold standard imaging tool to visualize brain anatomical structures.
3. Biopsy & Pathology: Surgical extraction of tissue sample to examine under a microscope for precise histological and molecular diagnosis.

Treatment Options

Treatment depends on tumor type, WHO grade, molecular biomarkers, size, and location:
- Surgery (Craniotomy): Maximum safe resection to remove as much tumor tissue as possible.
- Radiation Therapy: High-energy X-rays, Stereotactic Radiosurgery (Gamma Knife), or Proton Therapy targeting residual tumor cells.
- Chemotherapy: Oral or intravenous medications like Temozolomide (TMZ) that kill dividing tumor cells.
- Targeted & Tumor Treating Fields (TTFields): Specialized therapies targeting specific genetic mutations or electrical fields interrupting cell division.

Patient Support & Medical Advice Disclaimer

Multidisciplinary care involves neurosurgeons, neuro-oncologists, radiation oncologists, and oncology nurses. Patients and caregivers should actively participate in care decisions. IMPORTANT NOTICE: Information provided in this guide is for educational purposes only. Patients must consult qualified healthcare professionals for medical advice, diagnosis, and treatment planning.